Lecture 14: Section 4.1

Increasing and Decreasing Functions

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Increasing and Decreasing Functions

Definition (Increasing Function)

A function f is **increasing** on an interval if for any $x_1 < x_2$ in the interval, $f(x_1) < f(x_2)$.

Definition (Decreasing Function)

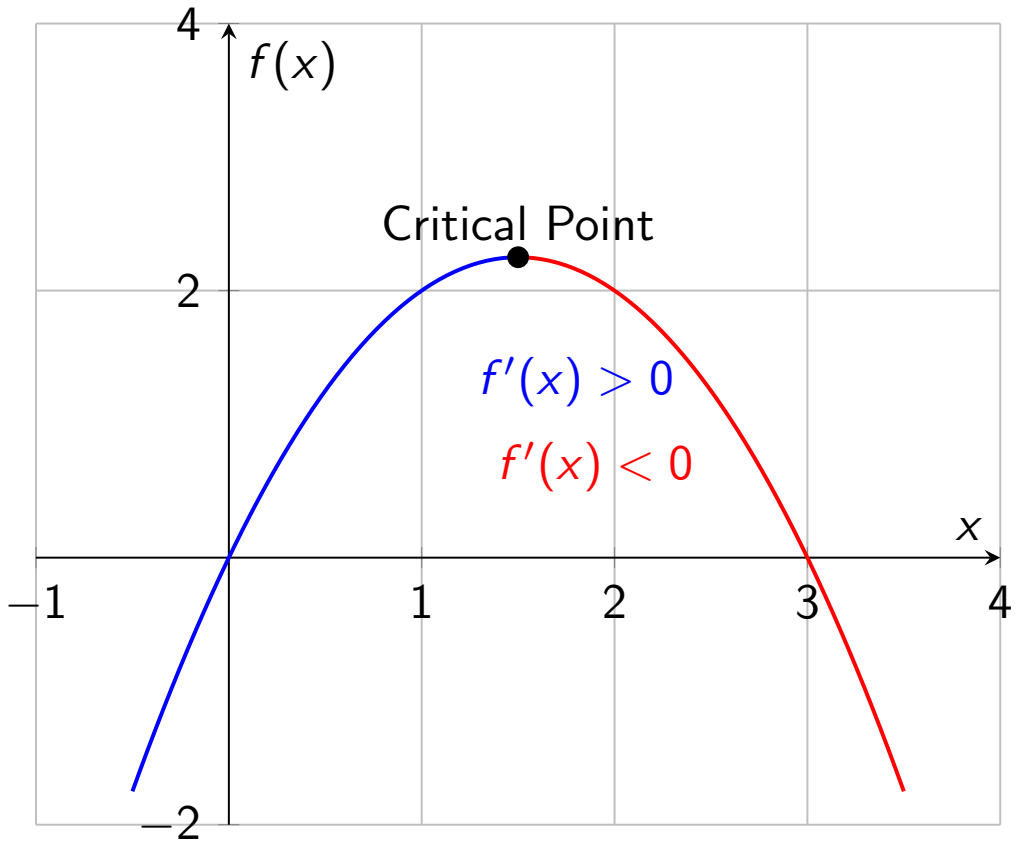
A function f is **decreasing** on an interval if for any $x_1 < x_2$ in the interval, $f(x_1) > f(x_2)$.

Theorem (First Derivative Test for Monotonicity)

Let f be differentiable on (a, b).

- *If $f'(x) > 0$ for all $x \in (a, b)$, then f is increasing on (a, b)*
- *If $f'(x) < 0$ for all $x \in (a, b)$, then f is decreasing on (a, b)*
- *If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b)*

Visualizing Increasing/Decreasing Functions



Example 1: Finding Intervals of Increase/Decrease

Find where $f(x) = x^3 - 3x^2 - 9x + 5$ is increasing/decreasing.

Solution:

1. Find derivative: $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3)$
2. Factor: $f'(x) = 3(x - 3)(x + 1)$
3. Critical points: $x = -1, 3$
4. Sign analysis:
   - $(-\infty, -1)$: Test $x = -2$, $f'(-2) = 3(-5)(-1) = 15 > 0 \Rightarrow$ Increasing
   - $(-1, 3)$: Test $x = 0$, $f'(0) = 3(-3)(1) = -9 < 0 \Rightarrow$ Decreasing
   - $(3, \infty)$: Test $x = 4$, $f'(4) = 3(1)(5) = 15 > 0 \Rightarrow$ Increasing

Answer: Increasing on $(-\infty, -1) \cup (3, \infty)$, Decreasing on $(-1, 3)$

Example 2: Trigonometric Function

Find intervals where $f(x) = \sin x + \cos x$ is increasing on $[0, 2\pi]$.

Solution:

1. $f'(x) = \cos x - \sin x$
2. Set $f'(x) = 0$: $\cos x - \sin x = 0 \Rightarrow \tan x = 1$
3. Critical points: $x = \frac{\pi}{4}, \frac{5\pi}{4}$
4. Sign analysis:
   - $(0, \frac{\pi}{4})$: Test $x = 0$, $f'(0) = 1 - 0 = 1 > 0 \Rightarrow$ Increasing
   - $(\frac{\pi}{4}, \frac{5\pi}{4})$: Test $x = \pi$, $f'(\pi) = -1 - 0 = -1 < 0 \Rightarrow$ Decreasing
   - $(\frac{5\pi}{4}, 2\pi)$: Test $x = \frac{3\pi}{2}$, $f'(\frac{3\pi}{2}) = 0 - (-1) = 1 > 0 \Rightarrow$ Increasing

Answer: Increasing on $(0, \frac{\pi}{4}) \cup (\frac{5\pi}{4}, 2\pi)$

Concavity

Definition (Concave Up)

A function is **concave up** on an interval if its graph lies above all its tangent lines on that interval.

Definition (Concave Down)

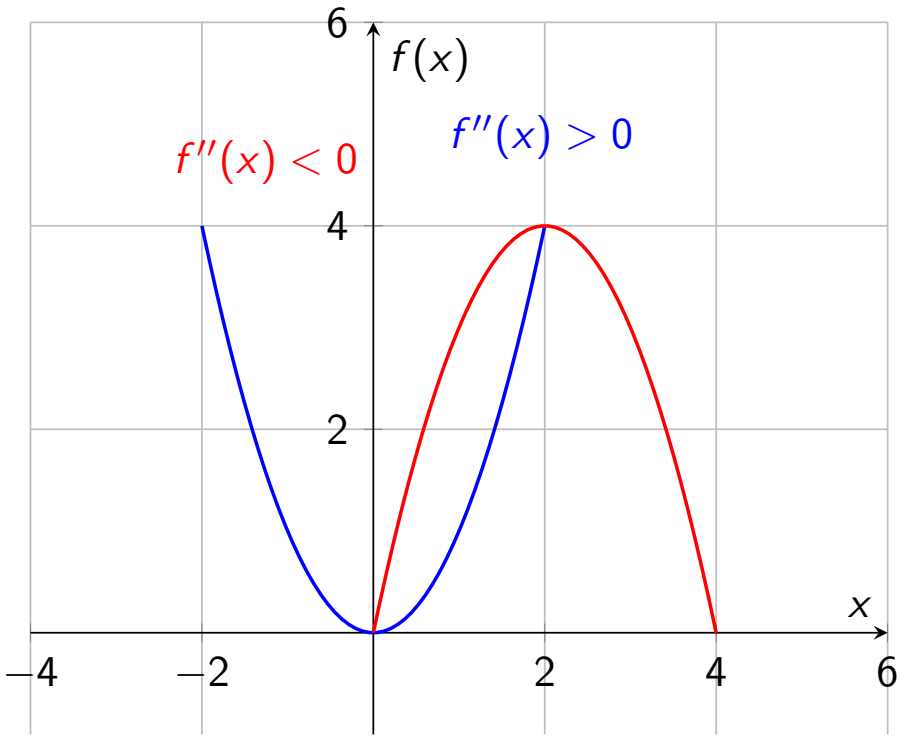
A function is **concave down** on an interval if its graph lies below all its tangent lines on that interval.

Theorem (Second Derivative Test for Concavity)

Let f be twice differentiable on (a, b).

- *If $f''(x) > 0$ for all $x \in (a, b)$, then f is concave up on (a, b)*
- *If $f''(x) < 0$ for all $x \in (a, b)$, then f is concave down on (a, b)*

Visualizing Concavity



Memory Aid:

- Concave Up: Holds water $\cup$
- Concave Down: Spills water $\cap$

Example 3: Finding Concavity

Determine concavity of $f(x) = x^4 - 4x^3 + 6x^2$.

Solution:

1. $f'(x) = 4x^3 - 12x^2 + 12x$
2. $f''(x) = 12x^2 - 24x + 12 = 12(x^2 - 2x + 1) = 12(x-1)^2$
3. Analyze $f''(x)$:
   - $f''(x) = 12(x-1)^2 \geq 0$ for all x
   - $f''(x) = 0$ only at $x = 1$
   - $f''(x) > 0$ for $x \neq 1$

Answer: Concave up on $(-\infty, 1) \cup (1, \infty)$

Note: Even though $f''(1) = 0$, the function is still concave up everywhere.

Example 4: Exponential Function Concavity

Analyze concavity of $f(x) = e^{-x^2}$.

Solution:

1. $f'(x) = -2xe^{-x^2}$
2. $f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = 2e^{-x^2}(2x^2 - 1)$
3. Since $e^{-x^2} > 0$ always, sign depends on $(2x^2 - 1)$
4. $f''(x) = 0$ when $2x^2 - 1 = 0 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$
5. Sign analysis:
   - $(-\infty, -\frac{1}{\sqrt{2}})$: $f''(x) > 0 \Rightarrow$ Concave up
   - $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: $f''(x) < 0 \Rightarrow$ Concave down
   - $(\frac{1}{\sqrt{2}}, \infty)$: $f''(x) > 0 \Rightarrow$ Concave up

Answer: Concave up on $(-\infty, -\frac{1}{\sqrt{2}}) \cup (\frac{1}{\sqrt{2}}, \infty)$, Concave down on $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

Inflection Points

Definition (Inflection Point)

An **inflection point** is a point on the graph where the concavity changes (from up to down or down to up).

Theorem (Finding Inflection Points)

If f has an inflection point at $x = c$, then either:

- $f''(c) = 0$, *or*
- $f''(c)$ *does not exist*

Important Note

$f''(c) = 0$ does NOT guarantee an inflection point! Must check that concavity actually changes.

Example 5: Finding Inflection Points

Find inflection points of $f(x) = x^3 - 3x^2 - 9x + 5$.

Solution:

1. From earlier: $f'(x) = 3x^2 - 6x - 9$, $f''(x) = 6x - 6$
2. Set $f''(x) = 0$: $6x - 6 = 0 \Rightarrow x = 1$
3. Check concavity change:
   - For $x < 1$: $f''(x) < 0 \Rightarrow$ Concave down
   - For $x > 1$: $f''(x) > 0 \Rightarrow$ Concave up
   - Concavity changes at $x = 1$!
4. Find y-coordinate: $f(1) = 1 - 3 - 9 + 5 = -6$

Answer: Inflection point at $(1, -6)$

Example 6: No Inflection Point Case

Show that $f(x) = x^4$ has no inflection point at $x = 0$.

Solution:

1. $f'(x) = 4x^3$, $f''(x) = 12x^2$
2. $f''(0) = 0$
3. Check concavity:
   - For $x < 0$: $f''(x) = 12x^2 > 0 \Rightarrow$ Concave up
   - For $x > 0$: $f''(x) = 12x^2 > 0 \Rightarrow$ Concave up
   - No change in concavity!

Answer: No inflection point at $x = 0$ even though $f''(0) = 0$

Example 7: Multiple Inflection Points

Find inflection points of $f(x) = \sin x$ on $[0, 2\pi]$.

Solution:

1. $f'(x) = \cos x$, $f''(x) = -\sin x$
2. Set $f''(x) = 0$: $-\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$
3. Check concavity changes:
   - $(0, \pi)$: Test $x = \frac{\pi}{2}$, $f''(\frac{\pi}{2}) = -1 < 0 \Rightarrow$ Concave down
   - $(\pi, 2\pi)$: Test $x = \frac{3\pi}{2}$, $f''(\frac{3\pi}{2}) = 1 > 0 \Rightarrow$ Concave up
   - Changes at $x = \pi$!
4. $f(\pi) = \sin \pi = 0$

Answer: Inflection point at $(\pi, 0)$

Note: $x = 0$ and $x = 2\pi$ are endpoints, not interior points.

Applications of Inflection Points

Inflection points represent important transitions in real-world contexts:

- **Economics:** Point of diminishing returns
- **Physics:** Maximum acceleration in motion
- **Biology:** Maximum growth rate in populations
- **Business:** Point of maximum efficiency gain
- **Engineering:** Point of maximum stress concentration

Example 8: Business Application

A company's profit function is $P(x) = -x^3 + 12x^2 + 60x$ (in thousands), where x is advertising spending (in thousands). Find the point of diminishing returns.

Solution:

1. Point of diminishing returns occurs at inflection point
2. $P'(x) = -3x^2 + 24x + 60$
3. $P''(x) = -6x + 24$
4. Set $P''(x) = 0$: $-6x + 24 = 0 \Rightarrow x = 4$
5. Check: $P''(x) > 0$ for $x < 4$, $P''(x) < 0$ for $x > 4$ ✓
6. $P(4) = -64 + 192 + 240 = 368$

Interpretation: Spending \$4,000 on advertising yields maximum marginal return. Beyond this, additional spending gives smaller profit increases.

Example 9: Physics Application

The position of a particle is $s(t) = t^3 - 6t^2 + 9t + 1$. Find when acceleration is maximum.

$$s(t) = t^3 - 6t^2 + 9t + 1$$
$$v(t) = s'(t) = 3t^2 - 12t + 9$$
$$a(t) = s''(t) = 6t - 12$$
$$j(t) = s'''(t) = 6$$

- Inflection points occur when $s''(t) = 0$ OR $s''(t)$ changes sign
- $s''(t) = 6t - 12 = 0$ when $t = 2$
- Check sign change:
  - For $t < 2$: $s''(t) < 0 \Rightarrow$ Concave down
  - For $t > 2$: $s''(t) > 0 \Rightarrow$ Concave up
  - Concavity changes at $t = 2$
- **Inflection point** of $s(t)$: $t = 2$ (where concavity changes)
- **Maximum acceleration**: No maximum on infinite domain

Final Answers:

- Acceleration has **no maximum** on $(-\infty, \infty)$
- Inflection point occurs at $t = 2$, position $s(2) = 3$

Logistic Growth Model

Definition (Logistic Function)

The **logistic function** models growth that is limited by carrying capacity:

$$P(t) = \frac{L}{1 + Ae^{-kt}}$$

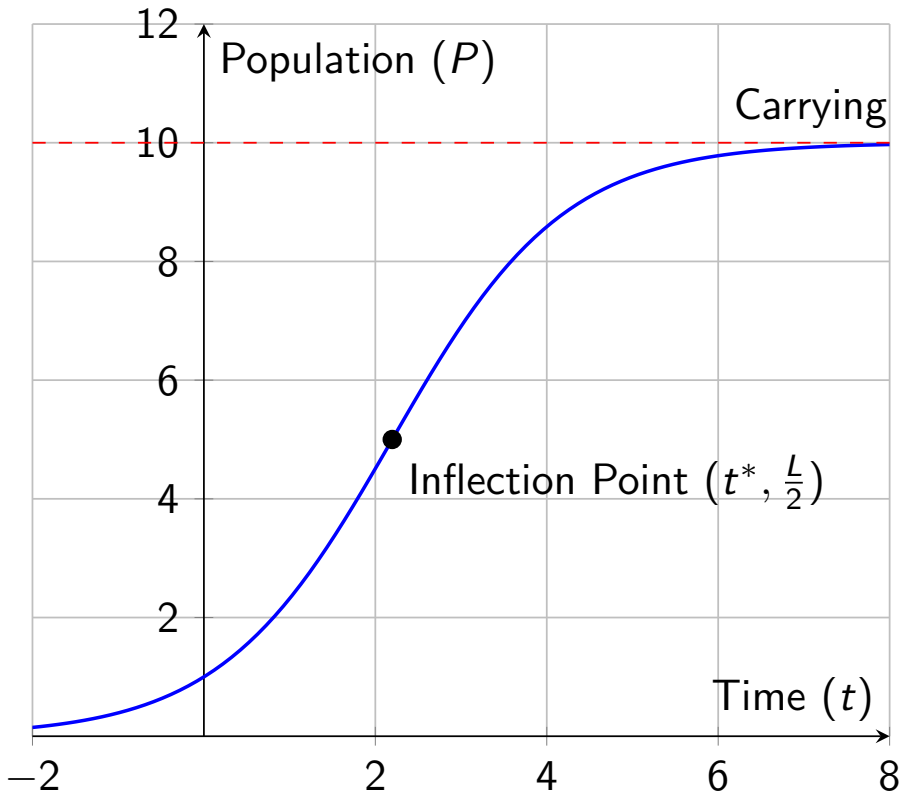
where:

- L = carrying capacity (maximum possible population)
- A = constant related to initial conditions
- k = growth rate constant
- t = time

Properties

- S-shaped curve (sigmoid)
- Initial exponential growth, then slows approaching carrying capacity
- Inflection point at half the carrying capacity

Logistic Curve Features



Example 10: Analyzing a Logistic Function

A population grows according to $P(t) = \frac{1000}{1+99e^{-0.5t}}$. Find:

1. Carrying capacity
2. Initial population
3. Inflection point
4. Maximum growth rate

Solution:

1. Carrying capacity: $L = 1000$
2. Initial population: $P(0) = \frac{1000}{1+99} = 10$
3. Inflection point occurs at $P = \frac{L}{2} = 500$
4. Solve $500 = \frac{1000}{1+99e^{-0.5t}} \Rightarrow 1 + 99e^{-0.5t} = 2 \Rightarrow t = \frac{\ln 99}{0.5} \approx 9.19$
5. Maximum growth rate occurs at inflection point

Answer: Carrying capacity $= 1000$, Initial $= 10$, Inflection at $t \approx 9.19$, $P = 500$

Practice Problems - Increasing/Decreasing

1. Find intervals where $f(x) = x^3 - 6x^2 + 9x + 1$ is increasing/decreasing.
2. Determine where $g(x) = \frac{x}{x^2+1}$ is increasing.
3. Find critical points and monotonicity of $h(x) = x^4 - 8x^2$.
4. Analyze $k(x) = \ln(x^2 + 1)$ for increasing/decreasing intervals.
5. For $m(x) = xe^{-x}$, find where it's increasing.

Practice Problems - Concavity & Inflection

1. Find concavity intervals for $f(x) = x^3 - 3x^2 - 9x + 5$.
2. Locate inflection points of $g(x) = x^4 - 4x^3 + 6x^2$.
3. Analyze concavity of $h(x) = \frac{x}{x^2+4}$.
4. Find all inflection points of $k(x) = \sin x + \cos x$ on $[0, 2\pi]$.
5. Determine concavity of $m(x) = e^x + e^{-x}$.

Practice Problems - Applications

1. A bacteria population follows $P(t) = \frac{500}{1+49e^{-0.8t}}$. Find when growth rate is maximum.
2. Revenue is $R(x) = -2x^3 + 15x^2 + 24x$. Find point of diminishing returns.
3. Position function $s(t) = t^3 - 9t^2 + 24t$. Find when acceleration is zero.
4. Profit $P(x) = x^3 - 12x^2 + 36x$. Find production level for maximum marginal profit.
5. Logistic curve $L(t) = \frac{200}{1+19e^{-0.4t}}$. Find inflection point and interpret.

Summary of Key Concepts

- **Increasing/Decreasing:** Use first derivative $f'(x)$
- **Concavity:** Use second derivative $f''(x)$
- **Inflection Points:** Where concavity changes
- **Logistic Growth:** S-shaped curve with carrying capacity
- **Applications:** Diminishing returns, maximum growth rate, etc.

Remember:

- $f'(x) = 0$ gives critical points
- $f''(x) = 0$ **may** give inflection points
- Always check sign changes!

Solutions - Increasing/Decreasing Problems

1. $f(x) = x^3 - 6x^2 + 9x + 1$
   - $f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x-1)(x-3)$
   - Critical points: $x = 1, 3$
   - Increasing: $(-\infty, 1) \cup (3, \infty)$
   - Decreasing: $(1, 3)$
2. $g(x) = \frac{x}{x^2+1}$
   - $g'(x) = \frac{(1)(x^2+1)-x(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$
   - Critical points: $x = -1, 1$
   - Increasing: $(-1, 1)$
   - Decreasing: $(-\infty, -1) \cup (1, \infty)$
3. $h(x) = x^4 - 8x^2$
   - $h'(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x-2)(x+2)$
   - Critical points: $x = -2, 0, 2$
   - Increasing: $(-2, 0) \cup (2, \infty)$
   - Decreasing: $(-\infty, -2) \cup (0, 2)$

Solutions - Increasing/Decreasing Problems (Cont.)

4. $k(x) = \ln(x^2 + 1)$
   - $k'(x) = \frac{2x}{x^2+1}$
   - Critical point: $x = 0$
   - Increasing: $(0, \infty)$
   - Decreasing: $(-\infty, 0)$
5. $m(x) = xe^{-x}$
   - $m'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x)$
   - Critical point: $x = 1$
   - Increasing: $(-\infty, 1)$
   - Decreasing: $(1, \infty)$

Solutions - Concavity & Inflection Problems

1. $f(x) = x^3 - 3x^2 - 9x + 5$
   - $f'(x) = 3x^2 - 6x - 9$, $f''(x) = 6x - 6$
   - $f''(x) = 0$ when $x = 1$
   - Concave up: $(1, \infty)$
   - Concave down: $(-\infty, 1)$
   - Inflection point: $(1, -6)$
2. $g(x) = x^4 - 4x^3 + 6x^2$
   - $g'(x) = 4x^3 - 12x^2 + 12x$, $g''(x) = 12x^2 - 24x + 12 = 12(x-1)^2$
   - $g''(x) \geq 0$ for all x, equals 0 at $x = 1$
   - No sign change in $g''(x)$
   - Concave up everywhere
   - No inflection points
3. $h(x) = \frac{x}{x^2+4}$
   - $h'(x) = \frac{4-x^2}{(x^2+4)^2}$, $h''(x) = \frac{2x(x^2-12)}{(x^2+4)^3}$
   - $h''(x) = 0$ when $x = 0, \pm 2\sqrt{3}$
   - Concave up: $(-2\sqrt{3}, 0) \cup (2\sqrt{3}, \infty)$
   - Concave down: $(-\infty, -2\sqrt{3}) \cup (0, 2\sqrt{3})$

Solutions - Concavity & Inflection Problems (Cont.)

4. $k(x) = \sin x + \cos x$ on $[0, 2\pi]$
   - $k'(x) = \cos x - \sin x$, $k''(x) = -\sin x - \cos x$
   - $k''(x) = 0 \Rightarrow -\sin x - \cos x = 0 \Rightarrow \tan x = -1$
   - Solutions in $[0, 2\pi]$: $x = \frac{3\pi}{4}, \frac{7\pi}{4}$
   - Check sign changes:
     - $(0, \frac{3\pi}{4})$: $k''(\pi/2) = -1 - 0 = -1 < 0$ (concave down)
     - $(\frac{3\pi}{4}, \frac{7\pi}{4})$: $k''(\pi) = 0 - (-1) = 1 > 0$ (concave up)
     - $(\frac{7\pi}{4}, 2\pi)$: $k''(\frac{15\pi}{8}) < 0$ (concave down)
   - Inflection points: $(\frac{3\pi}{4}, \sqrt{2})$, $(\frac{7\pi}{4}, -\sqrt{2})$
5. $m(x) = e^x + e^{-x}$
   - $m'(x) = e^x - e^{-x}$, $m''(x) = e^x + e^{-x} > 0$ for all x
   - Always concave up
   - No inflection points

Solutions - Application Problems

1. $P(t) = \frac{500}{1+49e^{-0.8t}}$
   - Maximum growth rate occurs at inflection point
   - For logistic function, inflection at $P = \frac{L}{2} = 250$
   - Solve $250 = \frac{500}{1+49e^{-0.8t}} \Rightarrow 1 + 49e^{-0.8t} = 2$
   - $49e^{-0.8t} = 1 \Rightarrow e^{-0.8t} = \frac{1}{49}$
   - $-0.8t = \ln(\frac{1}{49}) = -\ln 49 \Rightarrow t = \frac{\ln 49}{0.8} \approx 4.86$
2. $R(x) = -2x^3 + 15x^2 + 24x$
   - Diminishing returns at inflection point
   - $R'(x) = -6x^2 + 30x + 24$, $R''(x) = -12x + 30$
   - $R''(x) = 0$ when $x = 2.5$
   - Point of diminishing returns: $x = 2.5$
3. $s(t) = t^3 - 9t^2 + 24t$
   - $v(t) = 3t^2 - 18t + 24$, $a(t) = 6t - 18$
   - $a(t) = 0$ when $t = 3$
   - Acceleration is zero at $t = 3$ seconds

Solutions - Application Problems (Cont.)

4. $P(x) = x^3 - 12x^2 + 36x$
   - Marginal profit: $MP(x) = P'(x) = 3x^2 - 24x + 36$
   - Maximum marginal profit when $MP'(x) = 0$
   - $MP'(x) = 6x - 24 = 0 \Rightarrow x = 4$
   - Maximum marginal profit at production level $x = 4$
5. $L(t) = \frac{200}{1+19e^{-0.4t}}$
   - Carrying capacity $L = 200$
   - Inflection point at $L = \frac{200}{2} = 100$
   - Solve $100 = \frac{200}{1+19e^{-0.4t}} \Rightarrow 1 + 19e^{-0.4t} = 2$
   - $19e^{-0.4t} = 1 \Rightarrow e^{-0.4t} = \frac{1}{19}$
   - $-0.4t = \ln(\frac{1}{19}) = -\ln 19 \Rightarrow t = \frac{\ln 19}{0.4} \approx 7.36$
   - Interpretation: Maximum growth rate occurs at $t \approx 7.36$ when population $= 100$